



Personalized safety training for construction workers: A large language model-driven multi-agent framework integrated with knowledge graph reasoning

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ABSTRACT

Construction sites are inherently high-risk environments, making safety training for workers crucial to enhancing operational skills, reinforcing safety awareness, and reducing accident risks. Traditional centralized training often fails to engage workers due to monotonous nature and lack of relevance, leading to low efficiency. Moreover, critical resources such as operating instructions, safety guidelines, and accident reports are frequently mismanaged or underutilized. Therefore, this study proposes ConSTRAG, an innovative personalized construction safety training framework. By integrating large language model-empowered agents with knowledge graph reasoning, ConSTRAG generates tailored training materials, significantly improving the relevance and effectiveness of safety training. Validation tests conducted on a dataset of 11,020 questions achieved an average score of 81.25, exceeding the benchmark by 6.94. The generated personalized training materials were evaluated through an expert questionnaire survey, with an average score of 4.16 out of 5 across five dimensions. This research contributes to overcoming individual heterogeneity in construction safety training, enhances training efficiency and effectiveness, and holds potential for extension to other personnel training industries.

1. Introduction

The construction industry remains one of the most hazardous sectors globally (Chen et al., 2025; Chen and Yin, 2025; Yu et al., 2022), with worker safety being a critical concern due to high accident rates. Safety training is widely acknowledged as a key measure to mitigate occupational risks (Vignoli et al., 2021); however, approximately 80 % of construction accidents stem from human errors, including unconscious actions or misjudgments (Rasouli et al., 2025; Yoo et al., 2023). Although safety training serves as the primary method for knowledge dissemination (Xuecai et al., 2024), traditional lecture-based approaches dominate, often failing to engage workers effectively (Zhang et al., 2023). These methods suffer from inefficiencies such as inadequate personalization (Tong et al., 2023), fragmented knowledge delivery (Shahab et al., 2021), and limited long-term effectiveness, leaving workers unprepared to address dynamic operational risks. Furthermore, existing safety resources, such as standards and guidelines, are poorly

integrated into daily practices (Lee et al., 2024). This underscores an urgent demand for training frameworks that embed safety norms into operational workflows, bridging cognitive gaps and enhancing the practical value of safety knowledge.

Recent studies emphasize the need for personalized safety training addressing construction workers' heterogeneous roles and characteristics. For example, a complete construction project involves multiple tasks and activities that require different types of workers with distinct responsibilities. Signalmen guide lifting operations, riggers handle module loading and unloading, and crane operators manage heavy machinery (Zhang et al., 2023). These diverse roles demand tailored safety training to ensure accurate task performance (Cheng et al., 2023). Individual differences among workers, such as gender and age, can also influence the effectiveness of safety training (Loosemore and Malouf, 2019). Peiró et al. (2020) emphasize the critical role of individual trainee characteristics in safety knowledge transfer, while Lin et al. (2023) demonstrate that personalized interventions enhance training

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outcomes by addressing generic methods' limitations. Such personalized approaches reduce redundancy, enhance trainee satisfaction, optimize resource allocation, and maximize time efficiency (Tang et al., 2022). However, current training systems rely on conventional keyword-based retrieval methods, which suffer from ambiguity and poor contextual alignment (El-Gayar et al., 2019). Existing persona-based approaches, such as those proposed by Duan et al. (2024), remain static and misaligned with dynamic construction workflows. Additionally, time-intensive content organization further burdens workers (Muludi et al., 2024). Consequently, these limitations hinder the ability of training methods to address task-specific and environment-specific learning needs (Pedro et al., 2023).

Recent studies highlight the potential of large language models (LLMs) in construction management (Jiang et al., 2025; Wong et al., 2024). To address LLMs' domain knowledge limitations, researchers have developed retrieval-augmented generation (RAG) frameworks like LangChain (Topsakal and Akinci, 2023). By integrating external knowledge retrieval mechanisms, RAG enhances LLMs by mitigating hallucinations, improving interpretability, enabling adaptability, and reducing privacy risks (Han et al., 2024). However, applying RAG-LLM integration to personalized safety training faces a key challenge: developing agents that accurately retrieve domain-specific knowledge from unstructured construction management documents using limited worker-profile data (e.g., job roles, activity logs). Moreover, traditional RAG methods, such as text-chunk vector retrieval (i.e., NavieRAG (Gao et al., 2023)), exhibit limitations in retrieval precision (Uhm et al., 2025). Recently, graph-based RAG has emerged as a promising solution (Ludwig et al., 2025), demonstrating superior precision in knowledge retrieval. Knowledge graphs (KGs), serving as human-readable and machine-friendly knowledge representation models, facilitate reasoning and contextual understanding (Liu et al., 2025a; Simone et al., 2023; Yang et al., 2023). For example, studies by Bahr et al. (2025), Edge et al. (2024b), and Ludwig et al. (2025) highlight the effectiveness of graph-based approaches in improving retrieval performance. KG ontologies, which form the structural backbone of KGs, provide semantic, logical, and scalable frameworks for organizing domain knowledge. Nevertheless, the design of domain-specific ontologies tailored for construction management knowledge representation and retrieval remains underexplored.

Furthermore, prior research has primarily focused on safety training documents (i.e., regulatory standards and guidelines), such as the works of Pedro et al. (2023) and Duan et al. (2024). However, historical construction accident reports, which encompass basic accident information, causes, and lessons learned (Xuecai et al., 2024), are often underutilized despite their value as warning experiences in safety training (Tong et al., 2023). Fully leveraging these accident report databases to provide personalized safety warnings and accident case recommendations requires further exploration. This includes integrating rich contextual

details and ensuring relevance to individual worker profiles. Addressing these challenges is essential for developing a robust and effective personalized safety training framework that enhances training effectiveness, operational capability, and safety awareness among construction workers.

To address the aforementioned limitations, this study proposes a novel framework for personalized construction safety training, ConSTRAG, which integrates LLM-based agents with KG reasoning. The key idea of ConSTRAG, as illustrated in Fig. 1, is to achieve personalized construction safety training through three integrated components: (1) constructing safety training document databases alongside accident report databases via KG modeling and text vectorization; (2) generating tailored training content through worker-profile-driven knowledge retrieval; and (3) delivering personalized knowledge to workers via mobile applications. ConSTRAG extracts pertinent personalized knowledge from safety training document databases and accident report databases, with LLM-based agents integrated throughout the knowledge extraction and delivery pipeline to enhance process automation and accurately capture individual characteristics and preferences. This enables workers to acquire highly relevant safety knowledge and accident case experiences prior to engaging in specific construction tasks, thereby enhancing their operational competence and safety awareness. The main contributions of this study are outlined as follows:

- Proposing ConSTRAG for personalized construction knowledge retrieval and generation, which significantly improves knowledge retrieval accuracy through LLM-based agents and customizes the KG ontology for structured construction safety knowledge.
- Developing a construction management question bank (CMQB) containing 11,020 questions to evaluate ConSTRAG establishes a benchmark dataset for construction safety knowledge testing.
- Designs a prototype mobile application that delivers personalized safety training content tailored to individual workers' characteristics and work contexts, which enhances learning efficiency and training effectiveness of construction workers.

The rest of this paper is organized as follows. Section 2 provides a literature review summarizing safety training for construction workers and retrieval-augmented generation with KGs. Section 3 details the personalized construction safety training framework. Section 4 describes the databases and experimental design used to evaluate the proposed framework. Section 5 presents the experimental results, followed by a discussion in Section 6. Finally, Section 7 concludes the study with a summary of findings and suggestions for future research directions.

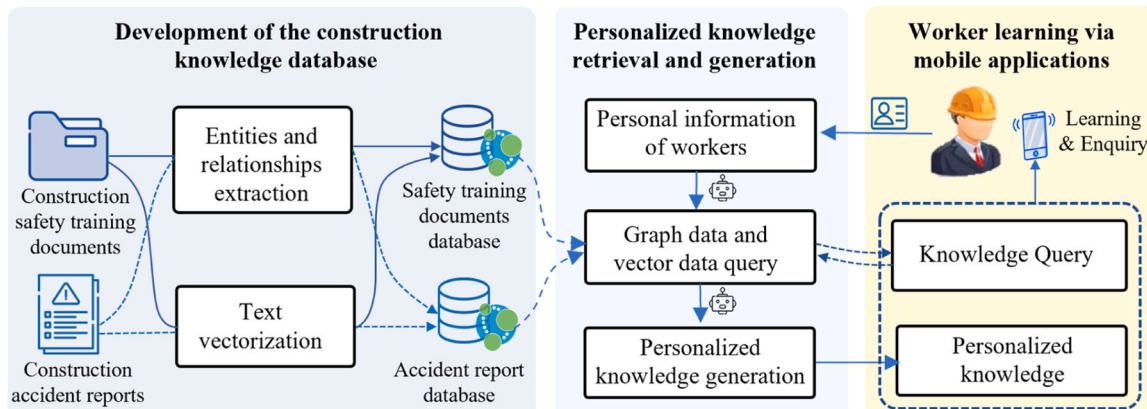


Fig. 1. The key idea of developing ConSTRAG for personalized construction safety training.

2. Literature review

2.1. Safety training of construction workers

Safety training is a cornerstone of effective safety management, playing a pivotal role in enhancing workers' safety performance and reducing the risk of injuries and fatalities (Loosemore and Malouf, 2019; Xiao et al., 2021a; Xuecai et al., 2024). The main goal of training workers is to help them acquire specialized operational skills and the ability to identify and respond to hazards. Numerous studies have shown that when workers receive adequate training, their safety performance improves, and the risk of safety incidents on construction sites is significantly reduced (Babalola et al., 2023; Xiao et al., 2021b). For example, a study by Sanmiquel et al. (2018) showed that providing workers with knowledge about the causes and factors of occupational safety and health hazards effectively enhances occupational safety performance. Rodríguez-Garzón et al. (2015) found that training is a major factor in improving the safety climate, safety perceptions, and safety behaviors in construction projects. Gao et al. (2019) confirmed the positive effect of training on improving construction safety through empirical research. Namian et al. (2022) demonstrated that training courses familiarizing workers with safety skills can help them perform their duties more efficiently and ergonomically, resulting in reduced fatigue levels. However, traditional training methods often require physical interaction and are associated with high operational costs, especially when such training needs to be repeated for large numbers of workers (Babalola et al., 2023). Traditional training methods also fail to equip trainees with the skills needed to analyze complex workplace situations and apply safety knowledge in practice (Vignoli et al., 2021). Additionally, frequent site rotations, unfamiliar work environments, and production pressures can hinder workers from participating in safety training (Fang et al., 2016).

According to the literature review by Lin et al. (2023), with the growing awareness of the importance of safety training, researchers have gradually explored different approaches to construction safety training and developed relevant tools using advanced technologies over the past decades. For example, Vignoli et al. (2021) proposed a safety training program for migrant workers, which is a theory-based program that can serve as a reliable and effective tool for organizations in the construction industry. Uddin et al. (2023) proposed the use of ChatGPT to help construction industry students improve their hazard identification skills during training. Guan et al. (2025) developed a web-based safety management platform for Chinese migrant workers in New Zealand, and the results showed that training effectiveness could be improved by providing localized safety policies and conducting multi-frequency, long-term safety training tests. In recent years, more studies have attempted to introduce virtual reality (VR) technology into worker training. For example, Hussain et al. (2024) developed a system, iSafeCom, that integrates multiple AI technologies (including LLMs) into VR simulations, offering an interactive, visual, and immersive training experience that meets the diverse needs of immigrant workers. Osti et al. (2021) explored the effectiveness of VR-based training on knowledge acquisition, task performance, self-efficacy, hazard recognition, and safety behaviors, among other metrics. Wolf et al. (2022) developed a framework for generating and evaluating trainee data in augmented virtual environments, and the test results showed that this framework could help improve safety training and assessment. Although these techniques show potential in worker safety training, existing VR systems still heavily rely on traditional computer input devices and hardware (Macias-Huerta et al., 2020), which may increase the cognitive load during the learning process. Additionally, these methods are often limited to specific construction safety scenarios, restricting the scope, breadth, and level of personalization in training.

Due to insufficient attention to the relationship between workers' individual characteristics, worker engagement is low and training effectiveness is poor (Duan et al., 2024). Developing personalized safety

training programs and providing materials that cater to different styles, needs, and expectations is particularly important (Xu et al., 2019). Xu et al. (2023) proposed a Bayesian Knowledge Tracing (BKT) model, which recommends personalized training materials based on trainees' learning progress. Validation results show that the BKT model performs well in imparting safety knowledge and reducing unsafe behaviors. Meanwhile, the individual characteristics of trainees may play an important role in the learning and transfer of safety training among migrant workers (Peiró et al., 2020). Cultural differences can also affect training outcomes, and a multicultural workforce presents several challenges for developing effective training programs, such as language barriers. Migrant workers often work with people from the same cultural background, which hinders their ability to learn the safety norms, regulations, and language of the host country (Peiró et al., 2020). Additionally, this issue may be further exacerbated if trainers lack the necessary knowledge and skills to provide effective safety training to a diverse workforce (Lin et al., 2018).

Another limitation is that, despite the importance of historical accident cases for conducting safety training (Xuecai et al., 2024), they are often neglected in the actual training process. Accident reports cover information such as basic accident details, causes of the accident, relationships between causes, violated regulations, and accident consequences. In safety management research, these data can serve as an objective source of textual information and are crucial for enhancing workers' safety awareness and preventing similar accidents in the future. Considering that learning from past experiences is a key step in improving safety, previous studies often focus on analyzing the macro patterns of construction accidents. For example, Huo et al. (2024) proposed topological indicators to explore the internal characteristics of multidimensional construction accident networks and developed prevention strategies for subway construction safety management. Similarly, Kumi et al. (2025) constructed a construction accident network to represent safety factors and their co-occurrence patterns, providing a more comprehensive approach to accident prevention. Additionally, Guo et al. (2022) demonstrated that accident reports enable the understanding of how construction accidents occur, thereby assisting and reinforcing workers' adherence to established safety protocols. Although valuable insights for training can be extracted from extensive construction accident data, directly applying accident cases to safety training is broadly acknowledged as effective (Tong et al., 2023). However, there remains a lack of sufficient research on how to effectively integrate these critical data into construction safety training.

2.2. Retrieval-augmented generation with KG

The rapid development of generative AI has driven the practical application and widespread implementation of LLMs, such as ChatGPT (Wu et al., 2023) and Qwen2.5 (Yang et al., 2024). These models have significantly improved language comprehension and generation by training on vast amounts of open-world textual data, enabling them to provide high-quality and diverse answers to almost any question. However, due to issues like hallucinations, LLMs often exhibit biases, inaccuracies, or outdated information (Ji et al., 2023; Wu et al., 2023). Additional challenges include high costs for knowledge updates and data security concerns. For instance, LLMs often fail to provide accurate answers regarding internal safety documentation within construction companies. RAG addresses these limitations by retrieving additional information from external data sources, thereby improving answer accuracy (Gao et al., 2023). A typical RAG framework comprises two key components: (1) a retrieval module that employs semantic search techniques to extract pertinent information from a knowledge database, where data are stored and indexed as embeddings in a unified vector space; and (2) a generation module that utilizes the retrieved domain-specific information as context for generating target content via the LLM. Lee et al. (2024) emphasize that the transparency and traceability of RAG architectures can enhance confidence among industry

practitioners. This approach is particularly beneficial for applications such as question answering, summarization, and translation. In recent years, RAG has been applied in the construction industry primarily in question-answering systems, such as defect Q&A systems (Jeon and Lee, 2025), contract Q&A for construction management (Taiwo et al., 2024), and safety information generation (Uhm et al., 2025).

However, traditional RAG approaches, e.g., NaiveRAG (Gao et al., 2023), based on text chunking often result in retrieved chunks that are homogenized and redundant (Uhm et al., 2025). This method fails to reveal intrinsic relationships between the chunks and limits the LLM's ability to perform deeper reasoning. An increasing number of studies propose using semantic ontologies and KG-based approaches for domain knowledge representation and reasoning. Recently, integrating RAG with graph-structured data has provided a more efficient way to retrieve information, as seen in methods like GraphRAG (Han et al., 2024), LightRAG (Guo et al., 2024), MiniRAG (Fan et al., 2025). Knowledge in KGs is represented in the form of triples (head entity, relation, tail entity) and naturally linked through overlapping entities (Liu et al., 2025b; Sun et al., 2022). Storing unstructured data in a graph-structured KG facilitates precise information extraction through advanced graph reasoning. Consequently, constructing entities within graphs enables more effective handling of complex, domain-specific knowledge relationships (Li and Starly, 2024). Lee et al. (2024) demonstrated that RAG models effectively retrieve accurate construction industry terminology and standards when integrated with KGs. Although KG-enhanced RAG methods show promising outcomes, systematic research on the accuracy and reliability of construction-domain KGs for personalized safety knowledge retrieval and generation remains limited. Moreover, while techniques for building KGs from structured data are relatively mature, extracting entities and relationships from unstructured text remains a tedious and complex task. Methods like AutoKG (Chen and Bertozzi, 2023) offer possibilities for automating KG construction. Nevertheless, designing knowledge ontologies and automatically constructing domain-specific KGs from unstructured data, such as construction specification texts and accident reports, remains challenging.

2.3. Research gaps and research objectives

As reviewed in the above subsections, limited efforts have been devoted to personalized worker safety training. The two significant research gaps have been identified in this research are presented as follows:

- Lack of an integrated framework for personalized safety training. Although existing research has explored the integration of advanced technologies such as VR into safety training, these approaches are often constrained by issues such as high cognitive load and limited personalization. Language barriers also present significant challenges, especially for multicultural and migrant workforces who may struggle to comprehend the host country's safety norms, regulations, and terminology. This problem is further compounded when trainers lack the skills to effectively address cultural and linguistic diversity. Furthermore, while historical accident reports contain valuable information, such as causes, violated regulations, and consequences, that could substantially improve hazard recognition and prevention capabilities, there is a notable absence of systematic methods for integrating such data into safety training programs. In addition, the design and development of applications that can implement an integrated personalized safety training framework remain underexplored. Therefore, there is a critical need for a comprehensive framework that addresses these challenges while catering to the specific needs of construction workers.
- Challenges in automatically generating personalized training materials. Existing studies have proposed methods to recommend safety training materials based on predefined tags or labels. However, these approaches often lack dynamism and fail to align with the daily

activities and evolving needs of construction workers. While RAG techniques offer promising solutions by retrieving relevant information and generating customized content, conventional RAG methods that rely on text chunking (e.g., NaiveRAG) often yield homogenized and redundant outputs. These limitations hinder the ability to provide precise and contextually relevant training materials. To bridge this gap, further research is needed to integrate domain-specific KGs and design LLM-based agents capable of processing unstructured data, such as accident reports and construction regulations. Moreover, a comprehensive question bank dedicated to construction safety management is currently unavailable for testing and benchmarking. To the best of the authors' knowledge, no existing framework sufficiently addresses the demand for dynamically generated, personalized training materials tailored to individual worker profiles.

To address the identified research gaps, this study aims to develop a specialized framework, ConSTRAG, which integrates LLM-based agents with KG reasoning to enable extensive and efficient personalized safety training for construction workers. The framework is designed to tailor training content to individual worker characteristics and learning needs by leveraging construction safety documents and accident reports. Additionally, a prototype mobile application will be developed to facilitate the practical implementation of the framework, making it accessible and user-friendly for workers in diverse settings. By enhancing worker engagement, improving hazard recognition skills, and elevating overall safety performance on construction sites, this research seeks to contribute to reduced accident rates and more effective safety management practices.

3. Methodology

Fig. 2 illustrates the overall framework of ConSTRAG, comprising three core components: construction safety training document database development, construction accident database development, and personalized knowledge retrieval and generation. The first component involves systematically collecting of safety-related construction training documents, including technical standards, operational manuals, and safety protocols. The textual content is extracted and segmented into chunks, followed by vectorization for storage in the chunk vector database (denoted as CRCV). An LLM-based agent automatically constructs a domain-specific KG from these text chunks, archived in the graph database (denoted as CRKG). The second component uses web crawlers to collect construction accident reports. Through LLM-based agent refinement processes involving filtering and critical information extraction, processed reports are vectorized and stored in the construction accident report vector database (denoted as CARV). Each processed report is further analyzed by the LLM-based agent to extract entities and relationships, thereby constructing individual knowledge subgraphs, which are integrated into the construction accident knowledge graph (denoted as CAKG). The third component initiates with acquiring worker profiles and environmental parameters (e.g., weather conditions). The LLM-based agent performs dual-type keyword extraction (direct and associative keywords) to enable cross-database retrieval. The LLM-based agent identifies highly relevant knowledge elements through semantic matching, adapts retrieved content to individual worker profiles via contextual alignment, and generates tailored assessment questions. The finalized personalized knowledge is delivered to workers via a mobile application. Detailed explanations of these components are provided in subsequent sections.

3.1. Construction safety training document database development

In order to construct the construction safety training document database, a composite database $RD = [CRCV, CRKG]$, which contains the CRCV and CRKG, is established using an agent-based end-to-end

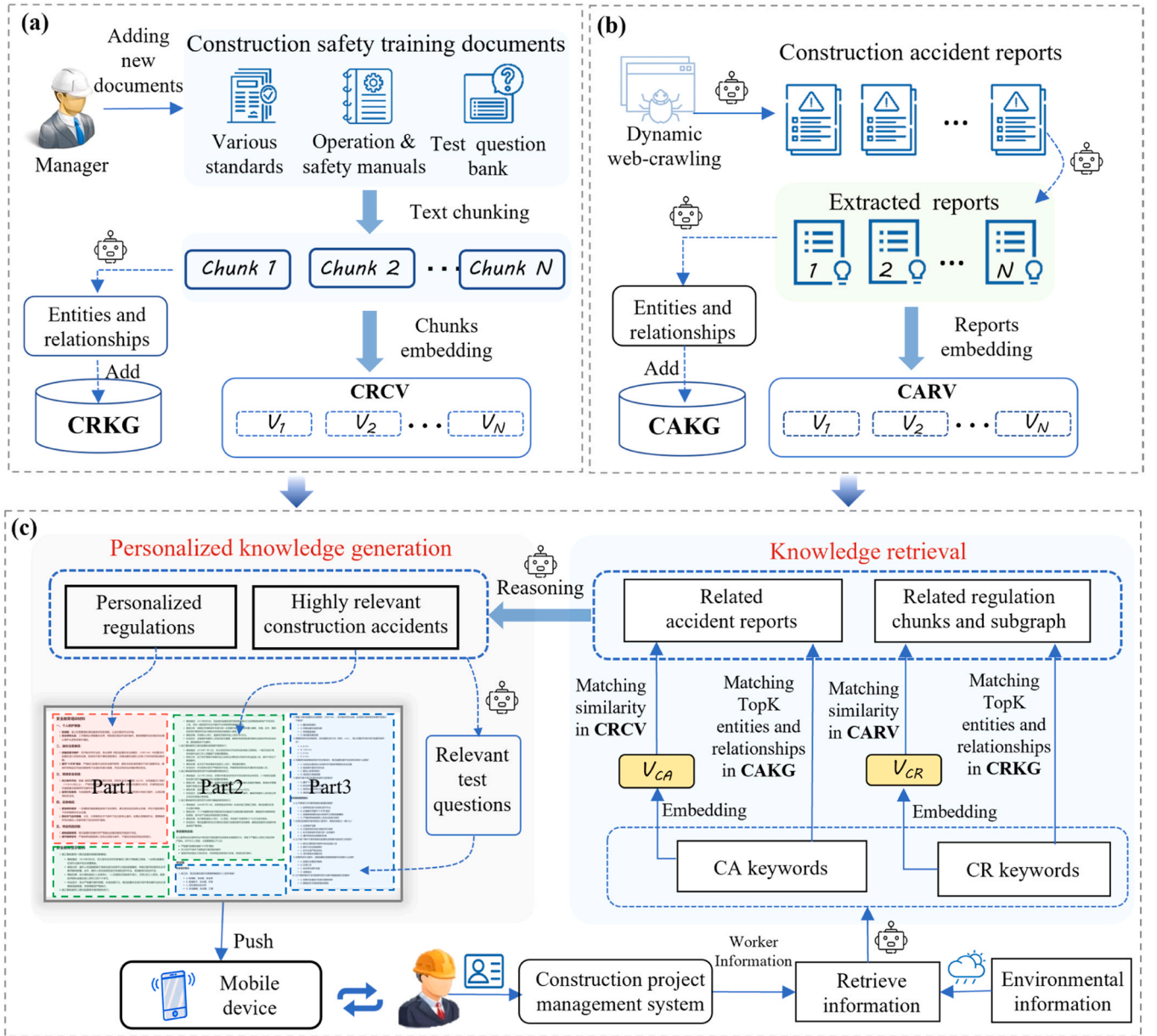


Fig. 2. Overall framework of ConSTRAG. (a) Development of the construction safety training document database, (b) construction of the accident report database, and (c) personalized knowledge retrieval and generation.

approach. The overall framework is shown in Fig. 3. Firstly, various types of safety training documents related to the construction industry are obtained through multi-source acquisition. Texts in image format are converted into plain text data through OCR VLM (Vision Language Model specifically designed for OCR tasks), forming the original text collection R_{raw} . On this basis, the text is segmented using the sliding window chunking strategy. By setting the overlapping region Δc between text chunks, both the semantic completeness of the text fragments and the avoidance of truncation at key information boundaries are ensured, ultimately generating an ordered text block sequence $C = [c_1, c_2, \dots, c_n]$. Subsequently, vectorized encoding is implemented for each text block $c_i \in C$, which is converted into dense vectors CV_i using a pre-trained text representation model. These vectors constitute the CRCV $= [CV_1, \dots, CV_n]$, providing underlying data support for subsequent applications.

Relationships between various entities provide complex details related to the construction process, safety regulations, and management measures. To construct the CRKG, this study refers to the research results

of Chen et al. (2024b) and Wu et al. (2021), combining with practical requirements to design the ontology O_r of the CRKG, as shown in Fig. 4 (a). The ontology contains 12 classes of entity types, with no strict relationship restrictions set between entities to adapt to diverse construction safety training documents. Each entity type covers all construction entities and their attribute data under the type, such as description information and chunk ID. For example, the entity “Asphalt heater” and its attributes are shown in Fig. 4(b).

The LLM-based agent is used to process c_i into a KG, and its design consists of five main steps: the first step involves the agent’s understanding of its role, explicitly indicating that the LLM needs to extract entities and relationships from the input construction regulation text content; in the second step, the agent understands the description of O_r , including the definition of each entity type and the explanation of corresponding entity relationships, enabling accurate matching of target entities and relationships during extraction; in the third step, the agent understands the output format requirements, such as outputting extracted entities and relationships in the specified JSON format,

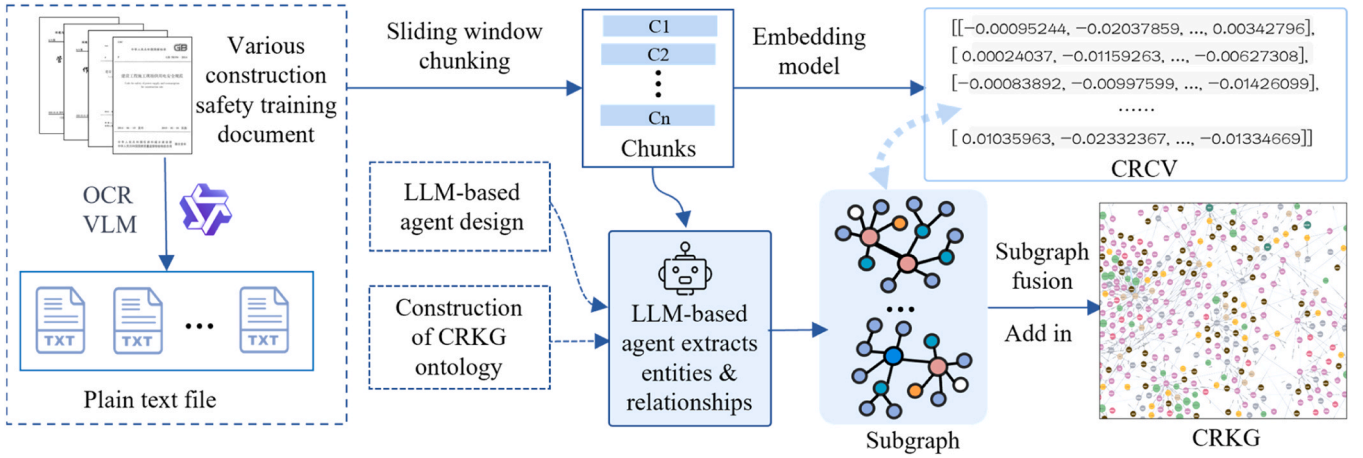


Fig. 3. End-to-end CRCV and CRKG construction process.

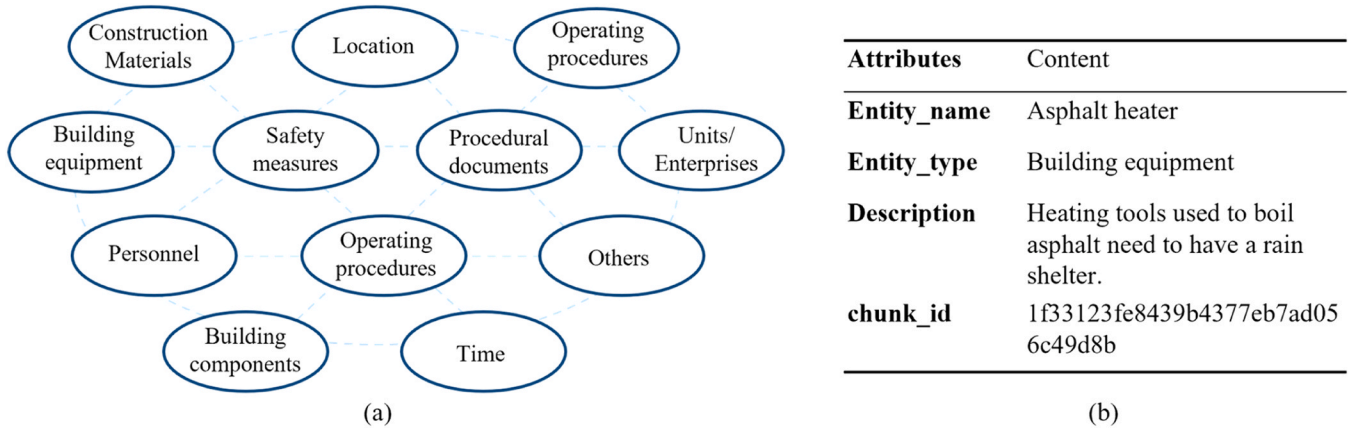


Fig. 4. CRKG ontology and its entity examples. (a) Ontology of CRKG and (b) the entity "Asphalt heater" and its attribute representation.

ensuring structured and parsable results; in the fourth step, the agent extracts content from the input text block c_i and outputs the result; in the fifth step, the agent checks whether the output meets the requirements and determines whether all entities and relationships have been extracted. If not, it corrects the output and continues extraction until the conditions are met. Through a multiple-round iteration mechanism, i.e., after initial extraction, the agent revisits the input text based on the extracted content and supplements missing entities and relationships to further improve KG construction. This mechanism enhances system robustness and significantly improves the comprehensiveness and accuracy of knowledge extraction. All reasoning and judgment functions are realized through prompt engineering. Interaction content is stored in temporary memory to support iterative optimization and reasoning judgment. The output of the agent is presented in triple form, constituting a CRKG subgraph. These subgraphs are associated with corresponding vector data in CRCV and integrated into CRKG through entity alignment. Therefore, CRKG not only reflects relationships between entities but also forms an effective mapping with underlying vector data and corresponding text chunks, supporting subsequent knowledge retrieval and reasoning.

3.2. Construction accident report database development

Given the critical value of construction accident reports in safety training, it is important to fully utilize these data for personalized training. A LLM-based agent is utilized to build the construction accident report database $CA = [CARV, CAKG]$. The overall end-to-end

construction process is shown in Fig. 5. Accident reports record the cause, occurrence, and final result of an accident in detail. These reports are usually easily accessible from government websites and social media platforms. To efficiently collect a large number of accident report texts, automated collection is performed with the help of web crawler technology, and the acquired accident reports are represented by A_{Raw} . Meanwhile, in the data processing stage, these accident reports need to be analyzed in depth to identify key information and eliminate irrelevant content or privacy-related data, ensuring the accuracy and compliance of the database. This key information includes detailed descriptions of how the accident occurred, the direct and indirect causes of the accident, the consequences resulting from the accident, the lessons learned, and the time of the accident. By defining and extracting key information from accident reports, the quality and usability of the data can be improved, providing more accurate support for subsequent safety education and accident prevention. The first step in building a CAKG is to design its ontology. Referring to the research experiences of Chen et al. (2023), Liu et al. (2023) and Kumi et al. (2025), and combining them with the actual needs of this study, the ontology O_a of the CAKG is constructed, as shown in Fig. 6. O_a contains eight entity types and their relationships, which cover the core elements of construction accidents, and each entity contains multiple attributes.

The design of the LLM-based agent is the key to processing this data, and this process differs from the process described in Section 3.1. The first step involves the agent establishing a clear understanding of its role and task objectives, specifically to filter R_{raw} (filtered data are denoted as A_f), extract key information from A_f , and further complete the extraction

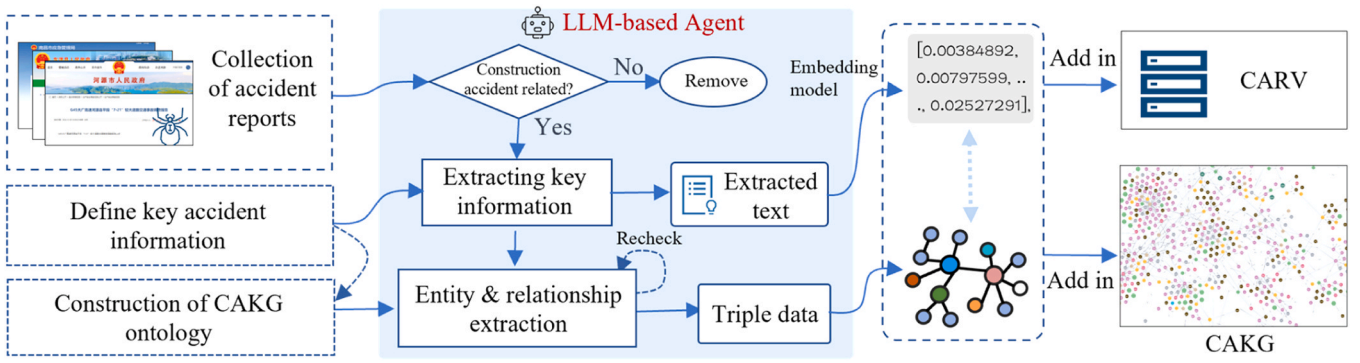


Fig. 5. End-to-end CAKG construction process.

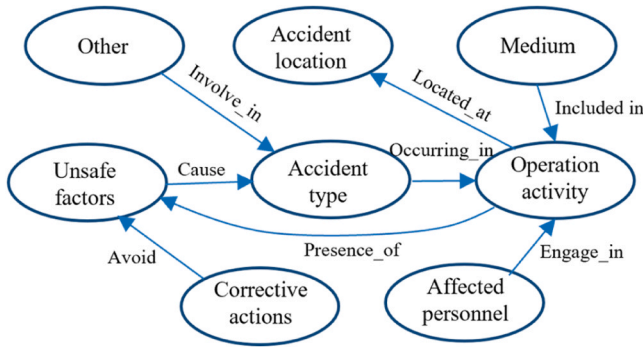


Fig. 6. Ontology of CAKG.

of entities and relationships. In the second step, the agent formally begins to filter crawled accident reports R_{raw} , comprehensively analyze report content, and determine whether it qualifies as a construction safety accident report. Non-relevant reports are discarded. In the third step, the agent needs to extract these core details from A_f , generating the purified construction accident report text A_e . In the fourth step, the agent familiarizes with the description of O_a , including the definitions of each entity type and the specific explanations of corresponding entity relationships. In the fifth step, the agent outputs the data in the specified JSON format to extract entities and relationships from A_e . In the sixth

step, the agent revisits the output data to determine whether it meets the specified output format requirements and confirms that all relevant entities and relationships have been completely extracted. If not, extraction continues until the requirements are met. This process makes full use of the agent's capabilities in end-to-end data processing, providing a reliable guarantee for building a high-quality CAKG. The information generated from each accident report constitutes a CAKG subgraph and is added to the CAKG through graph entity alignment. An example of the extracted A_e from an accident report and its corresponding subgraph is shown in Fig. 7. Meanwhile, each extracted construction accident text is converted into a one-dimensional vector representation using the text representation model and stored in CARV.

3.3. Personalized knowledge retrieval and generation

The personalized knowledge retrieval and generation framework is shown in Fig. 8. Two types of retrieval keywords are extracted, followed by obtaining relevant text materials through text similarity matching and KG reasoning. The personalized text is then generated through information refinement and personalized adaptation. The entire process fully utilizes the semantic understanding capability of LLM-based agent to generate personalized safety training materials.

3.3.1. Personalized knowledge retrieval

Personalized knowledge retrieval refers to making full use of personalized information to retrieve corresponding knowledge PK

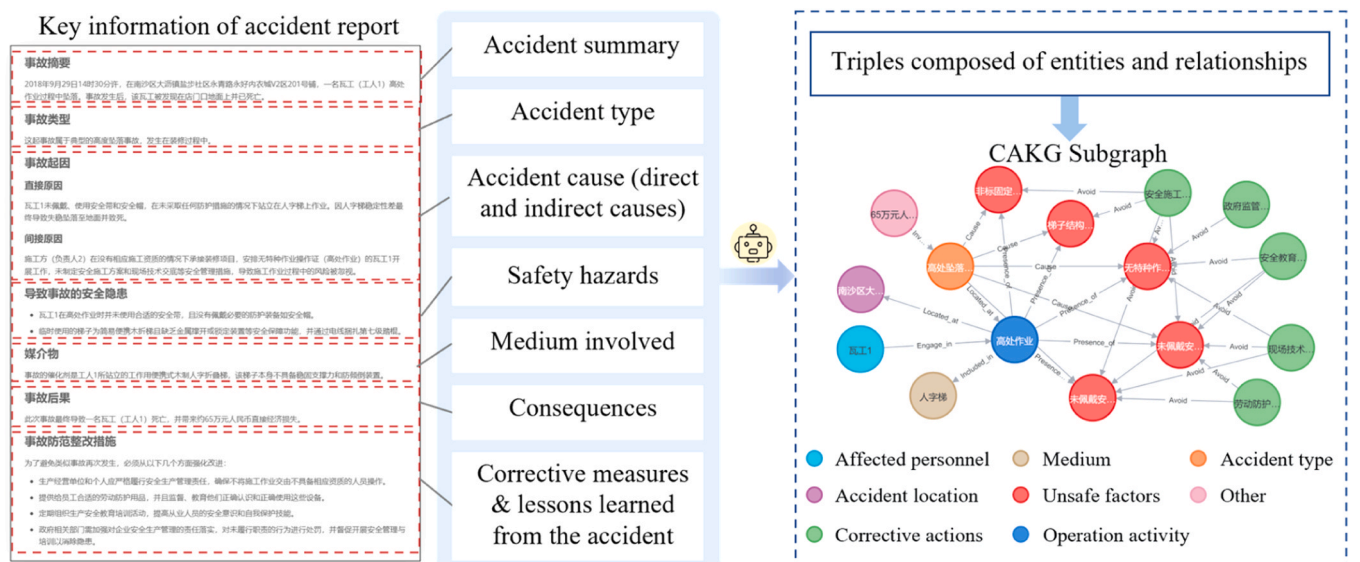


Fig. 7. Extracted key information from the construction accident report and corresponding subgraph.

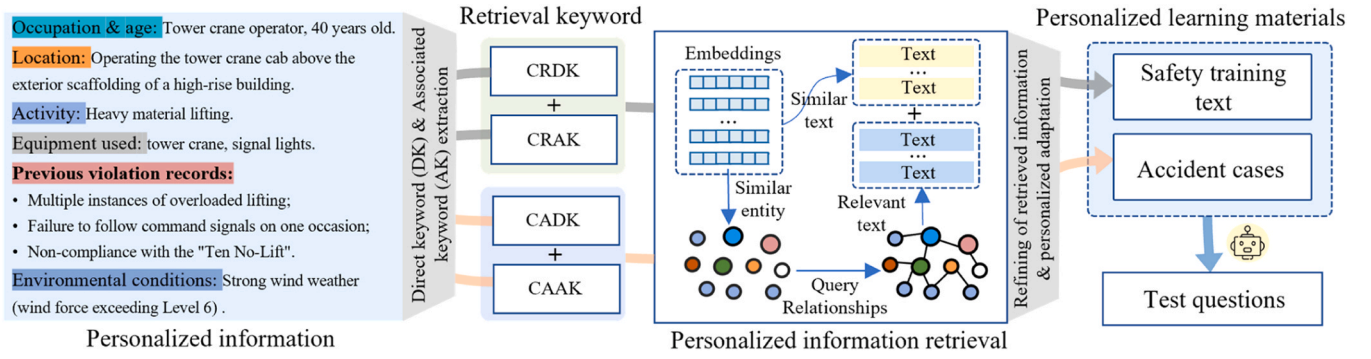


Fig. 8. Personalized knowledge retrieval and generation framework using an LLM-based agent.

$= [PRK, PAK]$ from CR and CA databases. As shown in the example in Fig. 8, personalized information includes occupation, age, location, equipment used, past violation records, and environmental information, which provide an important basis for accurately retrieving relevant knowledge. The personalized construction safety training document text retrieval process is illustrated in Algorithm 1. In the first line, LLM-based agent is used to generate construction regulation keywords (CRK), including direct keywords (CRDK) and associated keywords (CRAK). The processing flow of agent first requires understanding the task requirements and clarifying the format for keyword extraction and generation. Subsequently, the agent reads Q and generates $CRDK$ (all keywords directly related to Q , with no limit on the number of keywords) and $CRAK$ (generating M related keywords). Next, the agent rechecks the generated content for compliance until the conditions are met.

Algorithm 1. : Personalized regulation knowledge retrieval

Input: personalized information Q ; Database CR ; the number M of AK ; L_1 , L_2 and L_3 are the number of objects returned based on similarity.

Output: Retrieved personalized regulation knowledge PRK .

- 1 $CRK \leftarrow$ LLM-based Agent (Q, M) // $CRK = (CRDK, CRAK)$
- 2 $V_{CR} \leftarrow$ embedding (CRK)
- 3 $CRC_1 \leftarrow$ compute similarity ($CRCV, V_{CR}, L_1$)
- 4 $NS \leftarrow$ compute entity similarity ($CRKG, CRK, L_2$)
- 5 Subgraphs $RSG \leftarrow$ search edge by NS
- 6 $\vartheta \leftarrow$ calculate each entity degree in RSG
- 7 $CD \leftarrow$ get all Chunks id set in NS
- 8 $CRC_2 \leftarrow$ calculate chunk relevance (NS, ϑ, CD, L_3)
- 9 add CRC_1 and CRC_2 in PRK
- 10 **return** PRK

In the second line, CRK is converted into a string and represented as a vector V_{CR} using a text embedding model. In the third line, cosine similarity s_i is computed by comparing V_{CR} with each vector v_i in $CRCV$, as shown in the following equation:

$$s_i = \frac{V_{CR} \cdot v_i}{\|V_{CR}\| \|v_i\|} \quad (1)$$

Then, the L_1 vectors with the highest similarity are selected, resulting in the text CRC_1 retrieved from the vectors. In the fourth line, entity similarity is computed by comparing the vector representation of each keyword in CRK with the entities in $CRKG$ according to Eq. (1), and the L_2 entities with the highest similarity, denoted as NS , are returned. In the fifth line, corresponding subgraph RSG is obtained by searching through

the entities NS for all interconnected edges. In the sixth line, the degree of each entity in RSG is computed, and the degree ϑ_j of entity NS_j can be computed by the following formula:

$$\vartheta_j = \sum_{NS_i \in NS} A[NS_j][NS_i], \quad (2)$$

where $A[NS_j][NS_i] = 1$ indicates that there is an edge between NS_j and NS_i , otherwise $A[NS_j][NS_i] = 0$. In the seventh line, since each entity contains the chunk ID attribute, all chunks are grouped into a set CD . In the eighth step, the relevance CR of each chunk in CD is calculated. The relevance CR_k of the k -th chunk CD_k in CD can be computed by the following formula:

$$CR_k = \sum_{j=1}^n \vartheta_j \cdot 1_{CD_k \in NS_j}, \quad (3)$$

where n is the number of entities, and $CD_k \in NS_j$ means that CD_k is included in the set of chunk id attributes in NS_j . As a result, the L_3 chunks with the highest relevance are selected as CRC_2 . Finally, CRC_1 and CRC_2 are added to the set PRK . Similarly, PAK is obtained. Therefore, this fine-grained content retrieval method based on graph structure not only effectively integrates heterogeneous data from multiple sources but also significantly improves the accuracy of personalized knowledge retrieval through multi-level semantic analysis and association mining.

3.3.2. Personalized knowledge generation

PK at this point contains only information related to the Q , but this information still needs further refinement and organization to meet actual personalized requirements. This process is carried out by the LLM-based agent, as shown in Fig. 9. First, the agent fully understands the task requirements and then reasons about the input information Q and PK . The reasoning process includes generating text that meets the

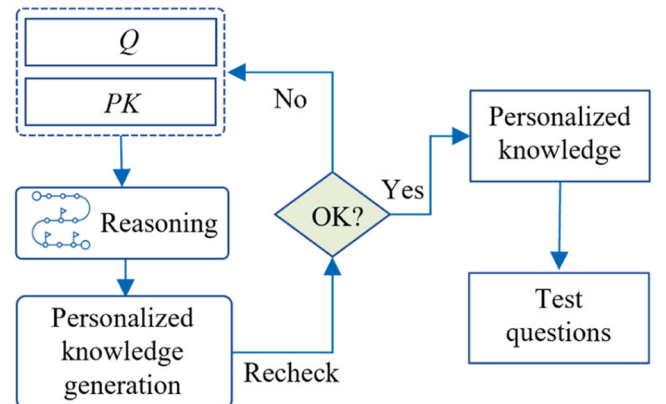


Fig. 9. LLM-based agent generates personalized knowledge process.

requirements of the regulation section of the designed training material and determining the number and content presentation format of the accident case section in the training material. This process does not involve the creation of new knowledge by the agent. During the generation process, the agent checks whether the output meets the requirements. If not, it iteratively optimizes the output until the conditions are satisfied. All conversations are saved as memory for the agent to support iterative optimization. Once the requirements are met, personalized training materials are finally generated. Testing plays an important role as an assessment tool in worker health and safety training (Weidner, 2000). To evaluate the learning outcomes of workers and reinforce their learning impression in practical applications, the LLM-based agent designs corresponding test questions based on the workers' personalized information and the generated training materials. Therefore, this method not only enhances the relevance and practicality of the training materials but also consolidates the learning outcomes of workers through testing, thereby providing more comprehensive support for worker training.

3.4. Mobile application development

The core of the personalized training approach is to provide users with a more convenient and user-friendly personalized learning experience through the learning and interaction process, leveraging the ConSTRAG framework proposed in this study. To achieve this goal, the mobile application design focuses on enhancing the user-friendliness and functionality of the interface, simplifying the learning process while comprehensively recording users' learning behavior and related information. The overall architecture of the application is shown in Fig. 10, which is divided into two main parts: the server side and the application side. On the server side, all tools are provided as APIs, including interfaces for retrieving workers' personal information and weather data, accessing graph and vector databases, and supporting various functions of the LLM-based agent. Among these, the LLM-based agent API plays a central role in the application, dynamically invoking other APIs based on user requests to perform data acquisition, processing, and integration, and ultimately delivering target information that supports highly personalized services. On the application side, the design emphasizes the UI interface and interaction logic. The UI interface includes key functional pages. Additionally, auxiliary features like content bookmarking, read-aloud functionality, and language conversion are incorporated to better accommodate users from diverse cultural and linguistic backgrounds, improving the application's accessibility and user experience. The interaction logic is designed to optimize navigation and operational flow between functional pages, ensuring seamless transitions between modules while maintaining intuitive and smooth interactions.

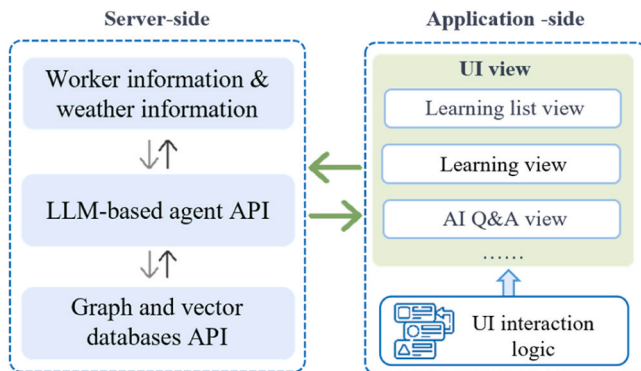


Fig. 10. Application software development framework.

4. Experimental design

4.1. Data collection and database construction

This study collects data from two main domains in China: construction safety training documents and construction accident reports. Regarding construction safety training documents, the safety standards and technical manuals comprise the *Technical Code for Safety of Working at Height in Construction*, *Safety Code for Electricity Supply and Use at Construction Sites of Construction Projects*, *Safety Regulations for Tower Cranes*, *Technical Code for the Safe Operation of Construction Machines*, and other construction training-related documents. These materials total 25 documents with a combined character count of 1,340,779. These documents provide rich foundational data support for subsequent KG construction and vectorization processing. Based on the method proposed in Section 3.1, we automatically construct the CRKG and the CRCV. Each text chunks are fixed at 1200 tokens, with Δc set to 200. In the vectorized representation process, the bge-m3 model (Chen et al., 2024a) is adopted, which efficiently transforms text content into semantic vectors and supports multiple retrieval methods, including dense retrieval, multi-vector retrieval, and sparse retrieval. With a maximum input length of 8192 tokens, this model fully satisfies the needs of long-text processing. Additionally, we select the Qwen2.5-32B (Yang et al., 2024) as the base LLM due to its robust capabilities, including support for inputs of up to 32,768 tokens, local deployment, efficient performance in handling complex tasks, and strong multilingual support, which collectively make it particularly well-suited for addressing diverse needs such as overcoming language barriers and ensuring accurate knowledge retrieval in multicultural environments. The final CRKG contains 15,453 nodes and 9789 edges, stored in the Neo4j graph database, while the CRCV includes vectorized representations of 1097 chunks.

A web crawler program was developed to collect raw construction accident reports from the safety management website (<https://www.safahoo.com/>), yielding a total of 9968 reports covering six common types of construction safety accidents, as summarized in Table 1. Following the methodology outlined in Section 3.2, an LLM-based agent was further designed to filter out reports unrelated to construction projects and extract key information. To evaluate the classification performance of the LLM-based agent, 300 randomly selected samples from each accident category were manually annotated using a custom-developed labeling tool. The classification performance for each type of accident report, illustrated in Fig. 11 (a), demonstrates that the precision for each accident type exceeded 0.75, while the recall approached nearly 1, indicating the model's high comprehensiveness in identifying positive samples and its ability to cover nearly all actual positive cases. Furthermore, the overall confusion matrix shown in Fig. 11 (b) reveals

Table 1

Counts of original and filtered construction accident reports.

Accident type	Original accident report count	Character count before filter	Filtered accident report count	Character count after extracted
A1: Struck-by object	1006	4368,295	558	480,163
A2: Vehicle accident	1048	6149,400	247	215,873
A3: Machinery accident	1015	4121,737	213	186,683
A4: Crane accident	1018	3454,181	662	574,798
A5: Electrical accident	1006	4139,442	585	503,640
A6: Fall from height	4875	18,543,070	3999	3374,195
Total	9968	40,776,125	6264	5335,352

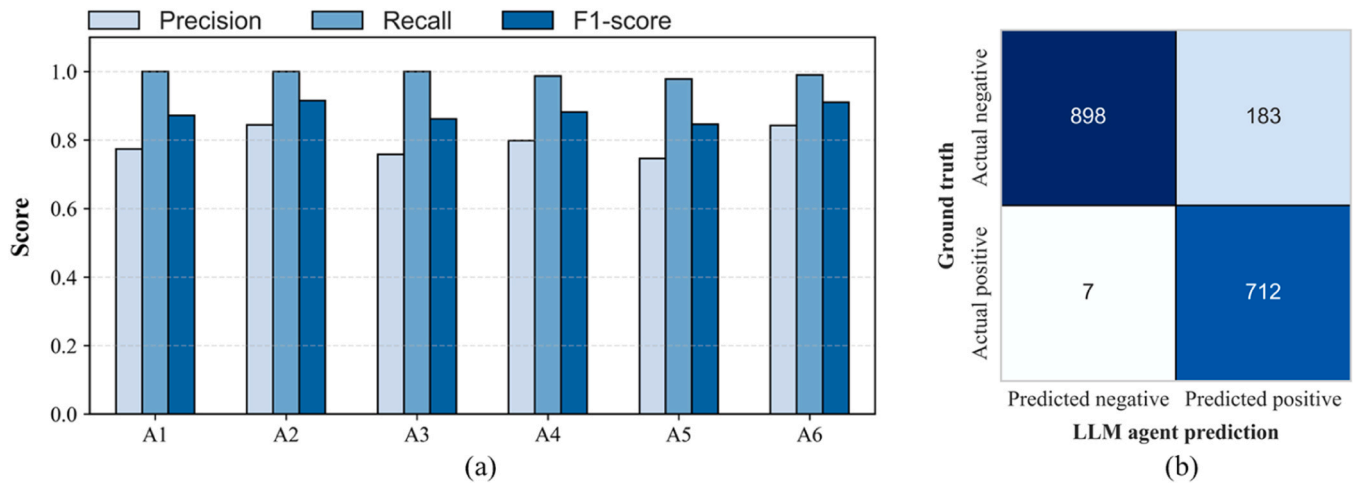


Fig. 11. Classification performance of the LLM-based agent. (a) Precision, recall, and F1-score for different accident types. (b) Confusion matrix of multi-class classification results.

strong positive sample prediction performance, suggesting minimal omission of positive samples and confirming the model's reliability in practical applications.

After processing, 6264 construction-related accident reports were obtained, with the "fall from height" category accounting for the largest proportion, reach 3999. Following information extraction, the total number of text characters was reduced to 5335,352, representing an 86.92 % reduction compared to the original data. This significantly reduces redundancy and irrelevant information. Subsequently, based on the processed data, the system automatically constructs a CAKG, containing 38,754 accident entities and 27,588 relationships, providing structured support for subsequent accident analysis and knowledge reasoning. Meanwhile, the CARV includes 6264 vectors, supporting efficient semantic retrieval and similarity calculations.

4.2. ConSTRAG performance evaluation experiment

In order to validate the effectiveness of the ConSTRAG proposed in this study compared to traditional methods, CMQB was constructed, containing 11,020 test questions generated from collected construction regulation documents. The specific process is as follows: special prompts were designed to enable the LLM to generate different types of questions, including multiple-choice, True/False, fill-in-the-blank, and short-answer questions, using chunks of text. Each question has a maximum score of 100 points. For outcome judgments, multiple-choice questions are in the form of four options, while True/False questions involve right-or-wrong judgments. These objective questions are directly assessed by the LLM for correctness, drawing on the studies of Guo et al. (2024) and Edge et al. (2024a). For fill-in-the-blank questions, scoring is based on the proportion of correct answers; for example, if only one of two blanks is answered correctly, the score is 50. Short-answer questions are evaluated based on three dimensions: Comprehensiveness, which measures how accurately the answer covers all aspects and details of the question; Empowerment, which assesses how well the answer helps the reader understand the question and enhances their correct understanding of the knowledge point; and Directness, which evaluates how specifically and clearly the answer addresses the question. The final evaluation result is the average score across these three dimensions, calculated using the locally deployed Qwen2.5-32B model. To further evaluate the performance of different models, three LLMs were selected for comparative experiments: the Qwen2.5 series (available in various parameter sizes), Llama3-8B, and Phi3-3.8B (both of which are highly regarded open-source models), as their diverse architectures and capabilities enable a comprehensive assessment of performance in CMQB. Meanwhile, to compare with the ConSTRAG, NavieRAG, i.e., the approach by

Uhm et al. (2025), was selected as the benchmark, which relies solely on data retrieved from vector databases (Gao et al., 2023) and does not incorporate other augmentation techniques. Both methods retrieve the same number of chunks, fixed at eight.

4.3. User experience experiment

Referring to past application development experiences, such as Nilsson (2009) and Zaina et al. (2022), and combining them with the software design framework proposed in Section 3.4, the software prototype for personalized safety knowledge training is shown in Fig. 12, which mainly consists of four key views. In my learning view, all learning-related knowledge (including learned, unlearned, and favorite content) is clearly displayed in a list, allowing users to quickly browse and manage their learning progress. In knowledge learning view, users can learn the personalized training knowledge. It also provides a test button to evaluate the learning effect and supports voice reading and content collection functions, thereby enhancing the convenience of learning and improving the user experience. In the AI Q&A view, users can input questions, and the software provides answers to help users address practical issues. In the profile view, users can view their learning records, personal information, and construction activity-related data, enabling them to fully understand their learning status and participation. The above page designs emphasize good visual effects and interactive experiences, aiming to provide users with an intuitive, efficient, and user-friendly operating environment, thus enhancing the application value of personalized safety training.

We systematically collected users' evaluation and feedback on the generated content of personalized safety training materials by designing and distributing a questionnaire based on a 5-point Likert scale. Specifically, construction management experts were invited to participate in the questionnaire survey. After reviewing multiple cases of personalized knowledge generation, the generated content was evaluated in five dimensions. These five dimensions include: Relevance, which refers to the extent to which the generated personalized knowledge matches the personal information of the workers, ensuring that the training content closely aligns with the actual background and needs of the workers; Accuracy, which measures the factual and technical accuracy and reliability of the generated personalized knowledge content. This includes whether the knowledge complies with current safety norms and standards, as well as its logical consistency and the absence of misleading information; Understandability, which focuses on whether the generated knowledge is presented in a clear, concise, and easy-to-understand manner; Learning effectiveness, which evaluates the extent to which workers' safety awareness, operational skills, and emergency

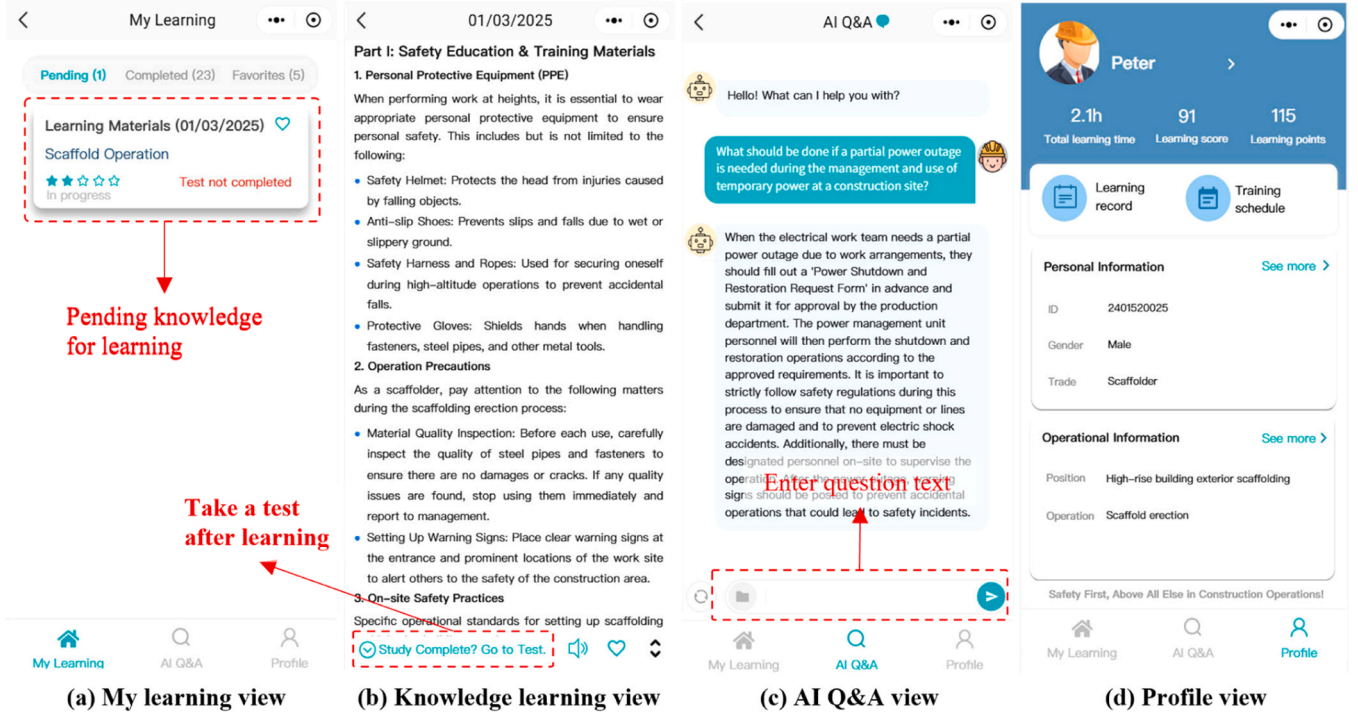


Fig. 12. Software prototype for personalized safety knowledge learning.

response capabilities are improved after learning the relevant content. This dimension focuses on whether the knowledge is effectively transformed into improved behavior and enhanced capabilities of workers; Management effectiveness, which assesses, from the perspective of construction management, whether these personalized training contents help improve the efficiency and safety of construction management. For example, it evaluates whether the content assists managers in optimizing construction processes or formulating more scientific safety management strategies, reflecting the value of personalized training at the organizational level. Through detailed evaluation of the above five dimensions, the quality of personalized safety training materials and their performance in practical applications can be fully understood, providing clear direction and a basis for subsequent optimization while laying a solid foundation for improving the overall level of construction safety training.

5. Results and analysis

5.1. ConSTRAG performance evaluation results

Table 2 show the performance of different LLMs on the CMQB. The Qwen2.5 series of models show a clear correlation between their

Table 2
Performance of different models on CMQB.

Models and parameter size	Types of test questions				Average score
	Selection	True/False	Fill-in-the-blank	Short-answer questions	
Qwen2.5-0.5B	17.19	17.42	16.13	43.02	23.44
Qwen2.5-1.5B	40.96	23.92	40.09	67.32	43.07
Qwen2.5-3B	75.28	75.54	28.96	72.99	63.19
Phi3-3.8B	62.00	58.68	6.94	36.58	41.05
Qwen2.5-7B	78.87	89.47	35.34	78.57	70.57
Llama3-8b	74.31	76.97	19.36	68.79	59.86
Qwen2.5-14B	84.01	88.36	42.70	79.06	73.53
Qwen2.5-32B	81.74	90.31	42.36	79.87	73.57

performance and parameter sizes when the model sizes are small. However, although the Qwen2.5-14B and Qwen2.5-32B models achieve the highest average scores (73.53 and 73.57, respectively), the rate of improvement slows down compared to smaller models, indicating that the performance of LLM in the construction management domain may be constrained by their inherent training data. The Qwen2.5-3B model outperforms the Phi3-3.8B model by 22.14, achieving a score of 63.19, while the Qwen2.5-7B model surpasses the Llama3-8B model by 10.71, reaching 70.57. These results suggest significant differences among the models in their ability to comprehend construction safety knowledge. From the perspective of question types, in objective questions, the accuracy of selection questions is higher, with the Qwen2.5-14B model achieving 84.01 on selection questions, while True/False questions reach a high score of 90.31 on the Qwen2.5-32B model. However, in subjective questions, fill-in-the-blank questions prove more challenging due to the need for precise understanding of the knowledge, with even the Qwen2.5-14B model only achieving a score of 42.70. As for short-answer questions, the Qwen2.5-32B model performs relatively well, achieving a score of 79.87. Thus, these findings highlight LLM deficiencies in handling detailed domain knowledge, particularly in tasks requiring high precision.

Table 3 shows the performance of NaiveRAG and ConSTRAG on LLMs with different parameter sizes. Overall, ConSTRAG outperforms NaiveRAG regardless of the LLM model's parameter size. For example, when using the Qwen2.5-1.5B model, ConSTRAG achieves an average score 7.8 higher than NaiveRAG. Additionally, as the number of LLM parameters increases, ConSTRAG's scores steadily improve, reaching a peak score of 81.25 on the Qwen2.5-32B model. However, NaiveRAG's scores decline for models with Qwen2.5-14B and Qwen2.5-32B parameters, achieving only 74.97 and 74.31, respectively. This may indicate that a method relying solely on vector retrieval has limitations when handling detailed tasks. Furthermore, in terms of scores for different question types, ConSTRAG's improvements are primarily reflected in subjective questions. For instance, when using the Qwen2.5-32B model, ConSTRAG scores 18.33 higher than NaiveRAG on short-answer questions, reaching 62.98, further demonstrating ConSTRAG's advantage in fine-grained knowledge retrieval and semantic

Table 3

Performance of NaiveRAG and ConSTRAG on LLM with different parameter sizes.

Qwen2.5 parameter size	Methods	Scores for different test question types				Average score
		Selection	True/False	Fill-in-the-blank	Short-answer questions	
0.5B	NaiveRAG	9.75	30.96	18.25	29.38	22.08
	ConSTRAG	12.30	24.15	18.82	37.18	23.11 (+1.03)
1.5B	NaiveRAG	14.13	14.18	63.21	55.17	36.67
	ConSTRAG	13.35	46.73	58.80	59.01	44.47 (+7.8)
3B	NaiveRAG	62.78	89.17	18.63	70.52	60.28
	ConSTRAG	62.10	83.78	32.25	74.83	63.24 (+2.96)
7B	NaiveRAG	72.23	88.81	31.54	74.18	66.69
	ConSTRAG	76.19	86.06	44.09	76.71	70.76 (+4.07)
14B	NaiveRAG	83.79	90.56	46.24	79.29	74.97
	ConSTRAG	83.48	91.80	58.82	80.72	78.7 (+3.74)
32B	NaiveRAG	83.67	91.11	44.65	77.82	74.31
	ConSTRAG	85.12	91.83	62.98	85.07	81.25 (+6.94)

understanding. Compared to the results in Table 2, ConSTRAG consistently outperforms the direct use of the LLM model, whereas NaiveRAG performs worse than the direct LLM model in cases with low parameter sizes (e.g., 0.5B and 1.5B). Even for models with Qwen2.5-14B and Qwen2.5-32B parameters, NaiveRAG only surpasses the LLM model by 1.44 and 0.56, respectively, which is significantly lower than ConSTRAG's performance (5.17 and 7.68, respectively). These results indicate that relying solely on vector retrieval is far less effective than ConSTRAG that incorporates KGs. ConSTRAG demonstrates the ability to capture detailed information more accurately through KG reasoning, significantly enhancing retrieval accuracy and application effectiveness for personalized knowledge retrieval.

5.2. Expert questionnaire evaluation results

To ensure practical relevance, the expert panel comprised scholars from university research institutes and construction professionals with extensive expertise in construction safety, ensuring a comprehensive understanding of safety training knowledge. The respondents' basic information is shown in Table 4. In terms of age distribution, the respondents are mainly concentrated in the 20–29 age group, accounting for 51.22 %. Regarding education, 73.17 % of the respondents hold a master's degree or higher (41.46 % with a master's degree and 31.71 % with a doctoral degree), indicating a high level of educational background and professionalism. Additionally, 46.34 % of the respondents have more than five years of work experience, further reflecting the professionalism and practical expertise of the sample. In terms of professional background, 48.78 % of the respondents are construction management experts from universities, 24.39 % are from research institutions, and 14.64 % are from construction companies. This diversity provides strong support for the professionalism and authority of the

Table 4

Demographic information of the respondents.

Category	Response	Frequency	Percentage (%)
Gender	Male	30	73.17
	Female	11	26.83
Age	20—29	21	51.22
	30—39	13	31.71
	40—49	3	7.32
	≥ 50	4	9.76
Educational level	Bachelor's degree or below	11	26.83
	Master's degree	17	41.46
	Doctoral degree	13	31.71
Work experience	≤ 5	22	53.66
	5 ~ 10	12	29.27
	≥ 10	7	17.07
Institution type	Universities	20	48.78
	Research institutes	10	24.39
	Construction companies	6	14.64
	Others	5	12.20

questionnaire results.

The study evaluates the effectiveness of personalized safety training materials from five dimensions using a five-level scale, and the specific results are shown in Table 5. The reliability of the sample responses was assessed using SPSS23.0 reliability analysis. The results of the analysis showed that the reliability coefficient value is 0.921, which exceeds 0.9, which exceeds the threshold of 0.8, thereby indicating high reliability according to standard assessment criteria.

As shown in Table 5, accident cases achieve the highest average score of 4.28, while test questions perform the lowest with an average score of 4.01. The overall performance evaluation reaches 4.19, indicating a high level of satisfaction among experts with ConSTRAG. The personalized knowledge generated by ConSTRAG achieves average scores exceeding 4.0 across all five dimensions. Specifically, relevance scores the highest at 4.25, demonstrating that the generated content aligns well with users' needs. This is followed by accuracy and learning effectiveness, with scores of 4.18 and 4.17, respectively, highlighting the significant role of personalized content in enhancing learning outcomes and supporting construction management. Understandability receives the lowest average score of 4.05, suggesting room for improvement in terms of clarity and ease of comprehension. The distribution of expert ratings across the five dimensions, evaluated on a five-level scale, is illustrated in Fig. 13. The number of ratings below 2 is fewer than ten for all dimensions, and the majority of ratings are concentrated between 4 and 5, indicating that experts generally evaluate these dimensions positively. Therefore, the personalized safety training materials demonstrate strong overall performance.

6. Discussion

The effectiveness of ConSTRAG was validated through testing on the constructed CMQB dataset and an expert questionnaire survey. The results of these experiments are discussed below to highlight key findings and provide guidance for practical applications.

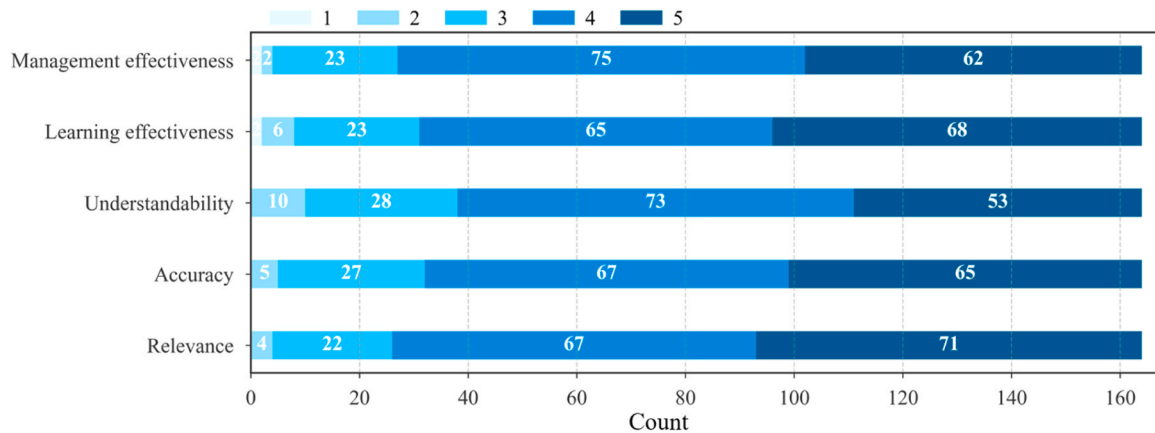
6.1. ConSTRAG performance evaluation

Differences exist in the performance of different LLMs and their parameter sizes regarding construction safety knowledge capabilities. As shown in Table 2, although the Phi3-3.8B model has more parameters than Qwen2.5-3B, its performance in the CMQB is 22.14 lower. Additionally, while the performance of the Qwen2.5 series improves steadily with increasing parameter size, it tends to plateau between Qwen2.5-14B and Qwen2.5-32B, with the latter being only 0.04 points higher. This phenomenon verifies that LLMs, despite being trained on large-scale corpora, may encounter bottlenecks in professional competence due to the limited availability of construction management-related knowledge in the training data (Heredia Álvaro and Barreda, 2025), such as the inability to use internal enterprise data for training model.

Table 5

Performance scores of ConSTRAG-generated personalized knowledge across five dimensions.

Content	Relevance	Accuracy	Understandability	Learning effectiveness	Management effectiveness	Average score
Regulation knowledge	4.17	4.22	3.95	4.32	4.05	4.14
Accident case	4.34	4.32	4.29	4.27	4.2	4.28
Test question	4.32	4.12	3.85	3.83	3.95	4.01
Overall performance	4.17	4.05	4.12	4.24	4.37	4.19
Average score	4.25	4.18	4.05	4.17	4.14	4.16

**Fig. 13.** Distribution of expert ratings for five dimensions evaluated on a five-level scale.

Therefore, to further enhance specialized capabilities, it is necessary to integrate domain-specific knowledge bases to supplement the capabilities of LLMs.

By combining LLM-based agents with KG, the effectiveness of personalized knowledge retrieval is significantly improved. As shown in Table 3, the performance of both NavieRAG and ConSTRAG improves with larger LLM sizes, but NavieRAG encounters a bottleneck and shows no significant improvement, even though Wong et al. (2024) argue that vector-based methods can enhance retrieval efficiency. In contrast, ConSTRAG's performance steadily increases, reaching 81.25 on Qwen2.5-32B, which is 6.94 points higher than NavieRAG. This improvement is particularly evident in fill-in-the-blank questions requiring fine-grained knowledge skills, suggesting that KG reasoning can effectively enhance fine-grained knowledge retrieval. Compared to the significant results achieved by NavieRAG used by Uhm et al. (2025) for construction management knowledge retrieval, the approach proposed in this study demonstrates superior performance, consistent with findings from Bahr et al. (2025), Edge et al. (2024b), and Guo et al. (2024).

LLM-based agents play an important role in the automated KG construction and personalized retrieval. This study constructs two types of KGs: CRKG and CAKG, differing from previous research methods. Most past studies relied on manual construction of KGs, such as Chen et al. (2023), who manually built CAKG, and Li and Starly (2024), who extracted key entities for CRKG by training a BERT model. In contrast, this study proposes an agent-based method for constructing domain-specific KGs, which does not require additional model training. Instead, it leverages LLMs' semantic understanding and task adherence capabilities with prompting techniques to enable end-to-end automated KG construction. This unsupervised approach reduces reliance on labeled data, enhancing framework scalability. During personalized knowledge queries, the LLM-based agent extracts two keyword types from worker profiles: DKs for precise knowledge location and AKs for retrieval breadth expansion. Throughout the process, it is evident that the LLM-based agent does not generate new knowledge but supports the application of the KG by processing existing knowledge (e.g., keyword extraction and knowledge integration). This approach, which does not

rely on the inherent expertise of LLMs or require fine-tuning, is more reliable, secure, and interpretable.

6.2. Expert questionnaire evaluation

The generated personalized training materials demonstrate high satisfaction and practical application value. As shown in Table 5, the professional questionnaire achieved an average score of 4.16 across five dimensions, indicating that the proposed method delivers valuable daily training materials highly relevant to user needs. For example, when using accident cases in safety training, it is crucial to ensure that the generated case information is both highly relevant and accurate. The construction safety accident case section of the generated materials scored 4.28, fully demonstrating the practical effectiveness of historical data in safety training. This reuse of historical experience effectively addresses the limitation of past studies that struggled to integrate such data with workers' actual needs. Meanwhile, the ConSTRAG-based software prototype effectively embeds personalized material recommendations into workers' workflows, enabling them to access task-relevant knowledge as conveniently as reading news headlines. This approach enhances workers' safety awareness and operational capabilities while fostering an overall safety culture. Managers need only update the knowledge base content, substantially reducing their workload. The results in Table 5 also show that learning effectiveness and management effectiveness scores reached 4.17 and 4.14, respectively, indicating significant results at both the individual learning and organizational management levels.

Construction safety training faces challenges from multicultural and multilingual work environments (Vignoli et al., 2021). The software ontology designed in this study effectively overcomes these limitations by considering workers' knowledge levels, cultural backgrounds, and native language usage habits. For instance, the system converts personalized materials into the target language and provides audio-based reading functions with accent adaptation, enhancing content comprehensibility and applicability. Moreover, the personalized training method proposed in this study is not limited to construction workers but can also be extended to construction managers and other

industries, such as chemical and transportation fields, showcasing broad application prospects.

6.3. Limitations

Based on the application practice of this study, the limitations, and potential improvements for enhancing personalized training effectiveness are outlined below. First, trainee knowledge levels should be incorporated into personalized knowledge generation. Feedback from the questionnaire survey indicated that test questions were overly simplistic, likely due to the high educational background of respondents (73.17 % held a master's degree or higher, as shown in Table 4). Adapting training materials to align with workers' knowledge levels would mitigate this bias and improve content relevance. Second, the construction training document database and accident report database developed in this study are constrained by limited data volume. A larger-scale domain-specific database covering diverse scenarios, activities, and conditions should be developed to enhance data comprehensiveness. Additionally, this study exclusively utilized text-based data, excluding multimodal sources (e.g., images and videos). Incorporating multimodal training data could improve engagement and effectiveness. Moreover, personalized materials should integrate real-time site-specific risk conditions (e.g., high-risk unsafe behaviors or job-specific hazards), aligning with Chen et al. (2023) approach to enhance practical relevance. Finally, extending ConSTRAG to support structured learning pathways tailored to topic-based content could significantly improve efficiency. By integrating adaptive syllabi into personalized programs, workers would benefit from systematic yet flexible learning experiences aligned with their roles and individual needs.

7. Conclusion

This study proposes a novel construction safety training framework, ConSTRAG, which enables personalized construction training knowledge generation by integrating LLM-based agents and KG reasoning. Unlike traditional methods that rely on trainer-led lectures or retrieval of training materials using tags and text vectors, ConSTRAG significantly enhances the efficiency of fine-grained knowledge structuring, personalized knowledge retrieval, and generation while considering the heterogeneity of workers. By leveraging construction training documents and historical accident reports, ConSTRAG demonstrates substantial advantages in automation, relevance, and adaptability for safety training. Furthermore, the framework shows potential to overcome language barriers faced by migrant workers, offering a worker-centered and efficient training approach. Future research will integrate hazard identification and risk assessment outcomes into ConSTRAG to enhance content relevance and practicality. Additionally, the framework will be extended to incorporate multimodal data (e.g., images and videos) for personalized training material generation, thereby enriching content presentation, improving user engagement, and supporting cross-industry applications.

CRediT authorship contribution statement

Qihua Chen: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Xianfei Yin:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Beifei Yuan:** Writing – review & editing, Investigation, Conceptualization. **Qirong Chen:** Resources, Investigation, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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